DATA VALUE CHAIN

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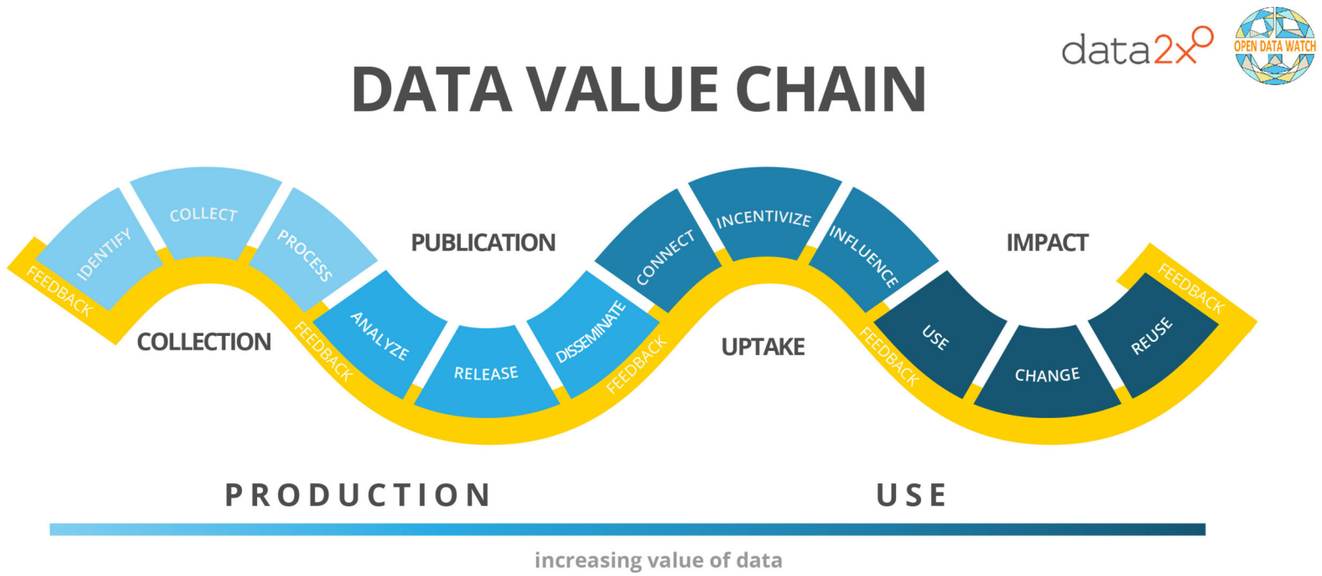
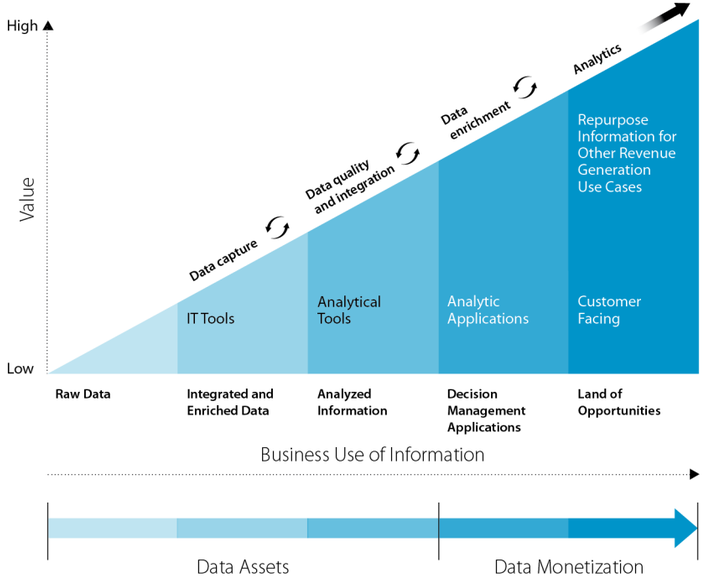
The Data Value Chain is a conceptual framework that outlines the various stages involved in transforming **raw data** into **valuable insights** and **actionable outcomes**. It emphasizes the processes and activities that organizations undertake to maximize the value derived from their data assets.

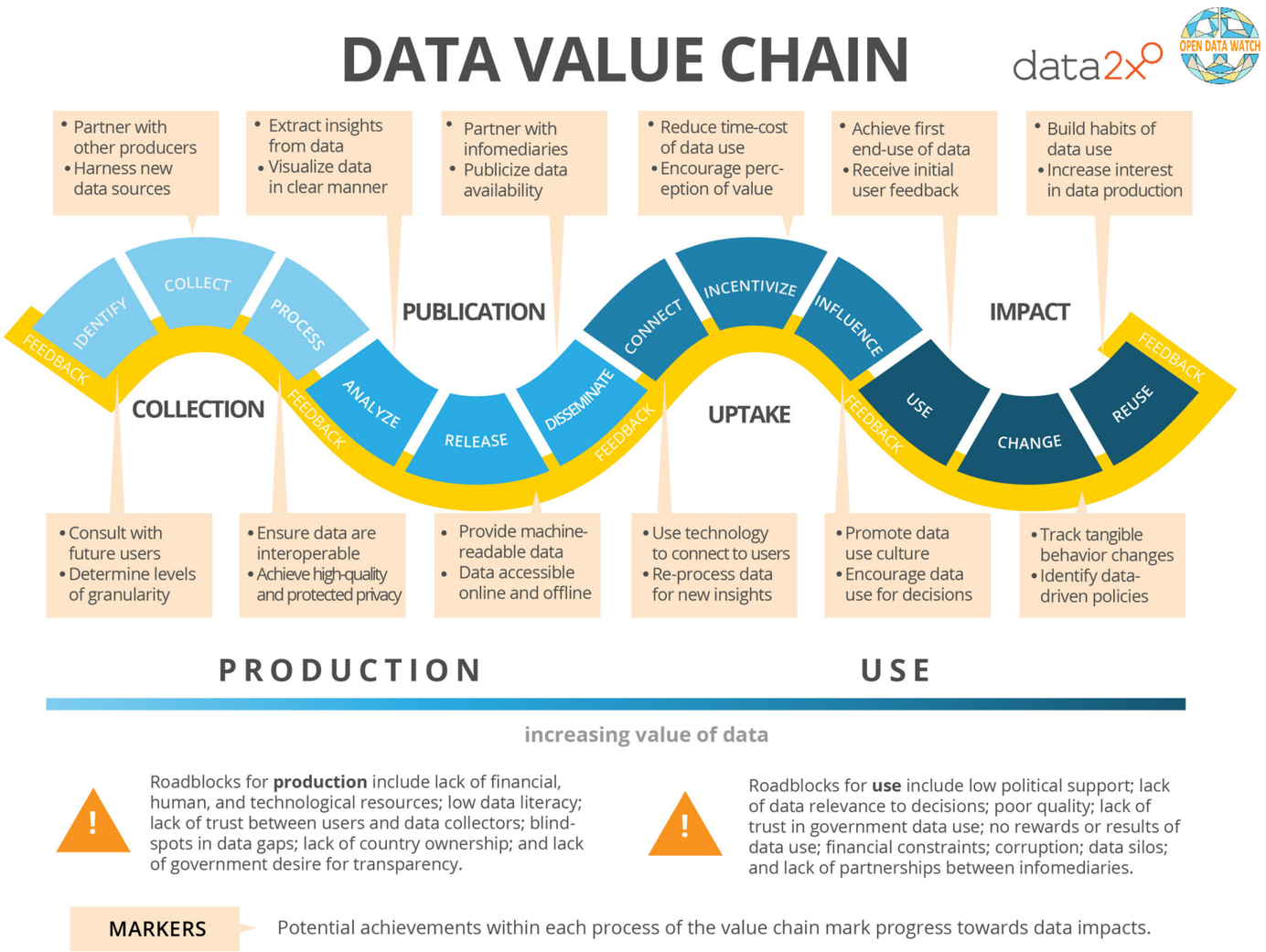


1. **Data Generation**: This is the initial stage where data is created or collected. Data can come from various sources, including sensors, transactions, social media, surveys, and more.
2. **Data Collection**: In this stage, the generated data is gathered and stored. This may involve data entry, data scraping, or automated data collection methods. The focus is on ensuring that the data is accurate, complete, and relevant.
3. **Data Storage**: Once collected, data needs to be stored in a way that allows for easy access and retrieval. This can involve databases, data lakes, or cloud storage solutions, depending on the volume and type of data.

4. **Data Processing**: This stage involves cleaning, transforming, and organizing the data to make it suitable for analysis. Data processing may include removing duplicates, handling missing values, and converting data into a standardized format.
5. **Data Analysis**: In this phase, analytical techniques are applied to the processed data to extract insights. This can involve statistical analysis, machine learning, data mining, and other analytical methods to identify patterns, trends, and correlations.
6. **Data Visualization**: The insights gained from data analysis are often presented through visualizations, such as charts, graphs, and dashboards. This helps stakeholders understand the findings and make informed decisions.
7. **Data Interpretation**: This stage involves interpreting the visualized data and insights to derive meaningful conclusions. It requires domain knowledge and expertise to understand the implications of the data.
8. **Data Utilization**: Finally, the insights are used to inform decision-making, drive strategy, improve processes, or create new products and services. This stage emphasizes the practical application of data insights to achieve business objectives.
9. **Feedback and Iteration**: The Data Value Chain is often iterative, meaning that feedback from the utilization stage can lead to new data generation, collection, and analysis efforts. This continuous cycle helps organizations adapt and improve their data strategies over time.



Recap


DATA VALUE CHAIN
data2x
OPEN DATA WATCH
• Partner with other producers
• Harness new data sources
• Extract insights from data
• Visualize data in clear manner
• Partner with infomediaries
• Publicize data availability
• Reduce time-cost of data use
• Encourage perc-eption of value
• Achieve first end-use of data
• Receive initial user feedback
• Build habits of data use
• Increase interest in data production
IDENTIFY
COLLECT
PROCESS
ANALYZE
RELEASE
DISSEMINATE
CONNECT
INCENTIVIZE
INFLUENCE
USE
CHANGE
REUSE
FEEDBACK
FEEDBACK
FEEDBACK
FEEDBACK
FEEDBACK
COLLECTION
PUBLICATION
UPTAKE
IMPACT
• Consult with future users
• Determine levels of granularity
• Ensure data are interoperable
• Achieve high-quality and protected privacy
• Provide machine-readable data
• Data accessible online and offline
• Use technology to connect to users
• Re-process data for new insights
• Promote data use culture
• Encourage data use for decisions
• Track tangible behavior changes
• Identify data-driven policies
PRODUCTION
USE
increasing value of data
Roadblocks for production include lack of financial, human, and technological resources; low data literacy; lack of trust between users and data collectors; blind-spots in data gaps; lack of country ownership; and lack of government desire for transparency.
Roadblocks for use include low political support; lack of data relevance to decisions; poor quality; lack of trust in government data use; no rewards or results of data use; financial constraints; corruption; data silos; and lack of partnerships between infomediaries.
MARKERS
Potential achievements within each process of the value chain mark progress towards data impacts.